

Record bonds and a mesh-pattern generating function

A proof of the conjecture for OEIS A289587

September 20, 2026

Abstract

We prove the conjectured ordinary generating function for the number of permutations avoiding the classical pattern 321 and either of the two mesh patterns designated (12, 174) and (12, 234). The crucial step is a structural characterization: in the first class, the mesh restriction is exactly the conjunction of having no adjacent consecutive ascent beginning at a left-to-right maximum and having at most one singleton direct-sum component. A canonical contraction of record runs reduces the enumeration to the Narayana refinement of 321-avoiding permutations. This gives the conjectured algebraic generating function, a refinement by three statistics, positive coefficient recurrences, an algorithm using linearly many integer arithmetic operations, and a two-term asymptotic expansion. The proof is self-contained at the enumerative level. Reproducible exhaustive and exact computations accompany the article.

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1 The problem and the result

Let a_n be the number of length- n permutations avoiding 321 and the mesh pattern (12, 174). We include the empty permutation, so $a_0 = 1$. OEIS A289587 identifies the same sequence with avoidance of (12, 234) and attributes the following generating-function conjecture to Thomas Scheuerle, December 23, 2025 [1]. The mesh-label convention is explained in Christian Sievers's December 2025 SeqFan discussion [2]; we specify the actual shaded sets below, so no encoding convention is left implicit.

Write

$$D(x) = x^4 - 2x^3 - 5x^2 - 2x + 1, \quad (1)$$

$$P(x) = x^4 + 4x^3 + 11x^2 + 10x + 3, \quad (2)$$

$$Q(x) = x^2 + 5x + 3. \quad (3)$$

Theorem 1.1 (The conjectured generating function). *For every $n \geq 0$, the two mesh-avoidance classes in A289587 are equinumerous. Their common ordinary generating function is*

$$A(x) = \sum_{n \geq 0} a_n x^n = \frac{P(x) - Q(x)\sqrt{D(x)}}{8x(1+x)^2}. \quad (4)$$

Here $\sqrt{D(x)}$ means the formal power-series square root with constant term 1. The apparent singularity at $x = 0$ is removable.

The first terms are

$$1, 1, 1, 3, 6, 18, 47, 139, 405, 1225, 3740, 11602, 36357, 115049, \dots$$

The interesting issue is not recognizing a square root from these numbers. A mesh condition quantifies over two selected entries and the emptiness of several surrounding rectangles. It can detect distant entries and therefore need not be reducible to a local prohibition. In the present case there is indeed an additional global obstruction: two singleton direct-sum components, even when separated by a nontrivial component. The proof isolates this obstruction before performing any generating-function manipulation.

The ambient class $\text{Av}(321)$ belongs to the central Catalan family of permutation enumeration. We derive the required Catalan–Narayana correspondence explicitly rather than using an unexplained formula for indecomposable permutations. The substantive new argument within this note is the mesh characterization and its connection to canonical record-run contraction.

Status and scope. The OEIS entry consulted on September 20, 2026 still explicitly labels (4) conjectural. This article supplies an independent proof of that formula; it does not assert an exhaustive bibliographic priority search. The two-pattern equivalence will also be proved, not inferred from agreement of initial terms.

2 Mesh patterns, records, and direct sums

2.1 The two shaded patterns

For a permutation $\pi = \pi_1 \cdots \pi_n$, a classical 321 occurrence is a triple $i < j < k$ with $\pi_i > \pi_j > \pi_k$. A 12 occurrence is a pair $i < j$ with $a = \pi_i < b = \pi_j$.

Relative to the latter pair, every other point (k, π_k) belongs to one of nine open rectangles. Its first cell coordinate $I \in \{0, 1, 2\}$ records whether k is before i , between i and j , or after j ; its second coordinate J records whether π_k is below a , between a and b , or above b . A mesh occurrence requires the rectangles in a specified set to be empty. This is the usual mesh-pattern framework [3], specialized here to a length-two pattern.

For the two labels in question, the shaded sets are

$$\mathcal{R}_{174} = \{(0, 1), (0, 2), (1, 0), (1, 2), (2, 1)\}, \quad (5)$$

$$\mathcal{R}_{234} = \{(0, 1), (1, 0), (1, 2), (2, 0), (2, 1)\}. \quad (6)$$

The convention in [2] assigns cell (I, J) the weight 2^{3I+J} ; summing the weights gives the displayed labels. These labels only name the patterns and play no arithmetic role in the proof.

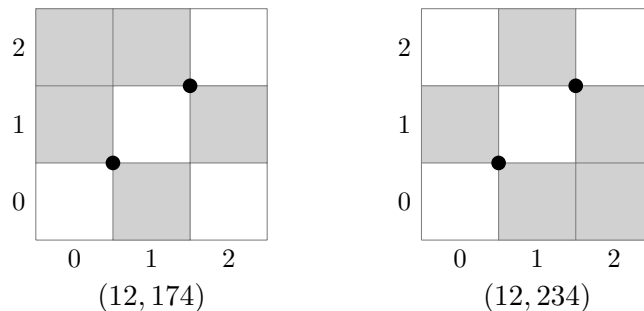


Figure 1: Gray cells must contain no other permutation points. Reflecting across the diagonal exchanges the two shaded sets. Outer cells represent unbounded rectangles, drawn here at finite size.

Lemma 2.1 (Inverse equivalence). *The map $\pi \mapsto \pi^{-1}$ is a bijection between the two classes counted in Theorem 1.1.*

Proof. Inverting a permutation transposes its point plot. The increasing pattern 12 is invariant under transposition, and transposition of every cell in (5) gives exactly (6). The classical pattern 321 is also its own inverse. Thus inversion preserves 321-avoidance and exchanges the two mesh restrictions, in both directions. $\square$

We henceforth work with $(12, 174)$. Refinements by records below refer to this class; the inverse bijection need not preserve the record statistic.

2.2 The structural statistics

An entry π_i is a *record*, or left-to-right maximum, when $\pi_h < \pi_i$ for every $h < i$. Write $\text{rec}(\pi)$ for the number of records.

Definition 2.2. An *upper bond* is an index $i < n$ such that π_i is a record and $\pi_{i+1} = \pi_i + 1$. A maximal chain of upper bonds is called a *record run*. A record not in any upper bond is a record run of length one.

The qualifier “upper” is important: we do not forbid every adjacent consecutive ascent. The ascent must start at a record.

For $\alpha \in S_p$ and $\beta \in S_q$, their direct sum is

$$\alpha \oplus \beta = \alpha_1 \cdots \alpha_p (p + \beta_1) \cdots (p + \beta_q).$$

A permutation is *sum-indecomposable* if it is nonempty and has no nontrivial such decomposition. A *sum cut* at t is characterized by

$$\{\pi_1, \dots, \pi_t\} = \{1, \dots, t\}.$$

Cutting at every sum cut gives the unique sequence of sum-indecomposable components. A singleton component means a component equal to the one-entry permutation 1. Equivalently, a point at position i is such a component when $\pi_i = i$ and the cuts immediately before and after that point are sum cuts. Boundary cuts at 0 and n are allowed.

3 The structural characterization

Theorem 3.1 (Local bonds and a global obstruction). *A 321-avoiding permutation contains (12,174) if and only if it has an upper bond or at least two singleton direct-sum components. Consequently, it avoids (12,174) exactly when neither obstruction occurs.*

Proof. Suppose first that positions $i < j$, with values $a = \pi_i < b = \pi_j$, form an occurrence of (12,174). The shaded rectangles say the following: all entries before i are less than a ; all entries strictly between i and j have values strictly between a and b ; and every value strictly between a and b occurs between i and j . The last assertion uses both the forbidden middle-valued prefix and the forbidden middle-valued suffix.

It follows that the position interval $[i, j]$ contains exactly the value interval $[a, b]$, with the endpoints fixed at a and b . In particular,

$$j - i = b - a, \quad a \geq i. \tag{7}$$

There are two different cases.

Case $a > i$. There are $a - 1$ values below a , but the prefix before i contains only $i - 1$ of them, and the block from i to j contains none. Therefore a value $c < a$ occurs after j . If the block from i to j had a descent $\pi_u > \pi_v$ with $u < v$, then $\pi_u > \pi_v > c$, at positions $u < v < \pi^{-1}(c)$, would form 321. The block is therefore increasing. Since its values are exactly $a, a + 1, \dots, b$, its first two entries are $a, a + 1$. The first is a record, so this is an upper bond.

Case $a = i$. The prefix before i is precisely the set $\{1, \dots, i - 1\}$. Equation (7) gives $j = b$. The initial j positions contain exactly $\{1, \dots, j\}$, and their last entry is j . Thus there are sum cuts at $i - 1, i, j - 1, j$, with repeated cuts identified when $j = i + 1$. The entries at i and j are two distinct singleton components. Notice that the middle block need not be increasing in this case.

Conversely, an upper bond $a, a + 1$ occupies adjacent positions. There are neither intermediate positions nor intermediate values, and every earlier entry is below a by the record condition. All five required rectangles are empty, so the bond is a mesh occurrence.

Finally, suppose there are two singleton components at positions $i < j$. Their values are i and j . Entries before i are below i , entries between the two components have values between i and j , and entries after j are above j . The selected singleton entries therefore form an occurrence of (12,174). This proves both implications. $\square$

Example 3.2 (Why both obstructions matter). The permutation 231 is excluded because 2,3 is an upper bond. The permutation $1324 = 1 \oplus 21 \oplus 1$ has no upper bond, but its two singleton components form a mesh occurrence. On the other hand, 4123 is allowed: its consecutive ascents 1,2 and 2,3 are not record ascents, and it is sum-indecomposable. Thus neither ordinary succession avoidance nor merely banning adjacent singleton components gives the right class.

Lemma 3.3 (Bonds at component boundaries). *An upper bond crossing a direct-sum boundary consists of two adjacent singleton components. Within a component, being a record is equivalent to being a record after standardizing that component.*

Proof. At a sum cut i , the largest value in the first i positions is i . If π_i is a record then $\pi_i = i$; if the next entry is consecutive, then $\pi_{i+1} = i + 1$. This forces additional sum cuts at $i - 1$ and $i + 1$, so both entries are singleton components. For the second assertion, all preceding components contain values smaller than every value of the current component. $\square$

Let $\mathcal{R}$ be the class of nontrivial (size at least two) sum-indecomposable 321-avoiding permutations with no upper bonds, and put

$$R(x) = \sum_{\pi \in \mathcal{R}} x^{|\pi|}. \quad (8)$$

The characterization and the boundary lemma give a unique decomposition of an allowed permutation: either a sequence of $\mathcal{R}$ -components, or a sequence of such components followed by one singleton and another sequence of such components. Therefore

$$A(x) = \frac{1}{1 - R(x)} + \frac{x}{(1 - R(x))^2}. \quad (9)$$

The second term counts an actual singleton component, not a marked arbitrary position. With m nontrivial components there are exactly $m + 1$ possible locations for it. Because $R(0) = 0$, these geometric-series constructions are legitimate formal power series.

4 Records in the Catalan class

To determine R , we first count all 321-avoiding permutations by records. Define

$$C(x, y) = \sum_{\pi \in \text{Av}(321)} x^{|\pi|} y^{\text{rec}(\pi)},$$

including the empty permutation with weight 1.

4.1 An explicit Dyck-path correspondence

Lemma 4.1. *In a 321-avoiding permutation, the subsequence of nonrecords is increasing. Conversely, a permutation whose nonrecords form an increasing subsequence avoids 321.*

Proof. If nonrecords $\pi_j > \pi_k$ occur at $j < k$, the fact that π_j is not a record supplies an earlier $\pi_i > \pi_j$, producing 321. Conversely, records themselves are increasing. If both records and nonrecords are increasing subsequences, any three entries contain two from the same subsequence, so they cannot be decreasing. $\square$

For a nonempty 321-avoider, list its record positions and values as

$$1 = p_1 < p_2 < \cdots < p_k, \quad v_1 < v_2 < \cdots < v_k = n.$$

Set $v_0 = 0$ and $p_{k+1} = n + 1$. Associate the path

$$U^{v_1 - v_0} D^{p_2 - p_1} U^{v_2 - v_1} D^{p_3 - p_2} \cdots U^{v_k - v_{k-1}} D^{p_{k+1} - p_k}, \quad (10)$$

where $U = (1, 1)$ and $D = (1, -1)$. The path has n steps of each kind, positive run lengths, and k peaks. Before the next record at p_{j+1} , the permutation has used $p_{j+1} - 1$ different values not exceeding v_j . Hence

$$v_j \geq p_{j+1} - 1. \quad (11)$$

This is exactly the assertion that the path remains nonnegative at the end of its j th descent run, and therefore everywhere.

Conversely, read the cumulative ascent lengths and descent lengths from a Dyck path. They recover the v_j and p_j in (10). Place the prescribed record values in their prescribed positions and fill all other positions increasingly with the remaining values. Inequality (11) guarantees enough unused values below v_j to fill every nonrecord position before p_{j+1} . Thus these fillings really are nonrecords, while each designated record exceeds all preceding values. Lemma 4.1 proves 321-avoidance. This is an inverse construction, establishing a bijection in which records correspond to peaks.

4.2 The functional equation and Narayana numbers

A nonempty Dyck path has its unique first-return form $U H D K$. When H is empty, the initial UD contributes one peak; otherwise the peaks are exactly those in H and K . Marking semilength by x and peaks by y gives

$$C = 1 + xyC + x(C - 1)C = 1 + xC(C + y - 1). \quad (12)$$

In particular $C(x, 1) = 1 + xC(x, 1)^2$, the Catalan generating function.

If $W = C - 1$, then $W = x(W + 1)(W + y)$. Lagrange inversion gives, for $n \geq 1$,

$$\begin{aligned} [x^n y^k]C &= \frac{1}{n} [t^{n-1} y^k] (t + 1)^n (t + y)^n \\ &= \frac{1}{n} \binom{n}{k} \binom{n}{k-1} =: N(n, k). \end{aligned} \quad (13)$$

We set $N(n, k) = 0$ outside $n \geq 1$, $1 \leq k \leq n$. These are the Narayana numbers; the derivation also establishes the precise record statistic being used. Standard formal inversion and later analytic transfer may be found in [4].

4.3 Removing the direct-sum decomposition

Let $I(x, y)$ count nontrivial sum-indecomposable 321-avoiders by size and records. Direct sum preserves 321-avoidance, and records are additive under direct sum. Each indecomposable component is either a singleton of weight xy or an object counted by I . Therefore

$$C = \frac{1}{1 - xy - I}.$$

Combining this with (12) yields the useful identity

$$I = 1 - \frac{1}{C} - xy = x(C - 1). \quad (14)$$

In particular $[x^n y^k]I = N(n - 1, k)$ for $n \geq 2$. Substituting $I = xW$ into the equation for W gives

$$I^2 + (x(1 + y) - 1)I + x^2 y = 0. \quad (15)$$

The desired solution is the one with zero constant term.

5 Canonical contraction of record runs

We now refine (8) by records:

$$R(x, y) = \sum_{\pi \in \mathcal{R}} x^{|\pi|} y^{\text{rec}(\pi)}, \quad R(x) = R(x, 1). \quad (16)$$

Theorem 5.1 (Record-run contraction). *Every nontrivial sum-indecomposable 321-avoiding permutation is uniquely obtained from a member of $\mathcal{R}$ by replacing each record with a nonempty increasing interval of consecutive values. Nonrecord entries remain single. All record-run lengths may be chosen independently.*

Proof. Start with such an indecomposable permutation π . Partition its records into maximal record runs, each occupying consecutive positions and carrying consecutive increasing values. Treat each nonrecord as a one-point block. Every block is therefore both a position interval and a value interval. Contract the blocks and standardize their values to obtain a permutation σ .

Avoidance and record status. Selecting one representative from each block gives σ as a classical pattern of π , so it avoids 321. The value intervals of different blocks are disjoint and totally ordered. A block is a record block exactly when all preceding blocks have smaller value intervals. A nonrecord's earlier larger witness lies in a different block and survives contraction. Thus records and nonrecords are preserved at the block level.

No new upper bond. Suppose two adjacent entries of σ formed an upper bond. Both entries would be records. Their original blocks are adjacent in position, and consecutive block ranks mean that no value block lies between their value intervals. Those intervals must therefore be consecutive as integer intervals. Their concatenation is a longer increasing record run in π , contradicting maximality. So σ has no upper bond.

Indecomposability of the contraction. A sum cut between contracted blocks would lift to the corresponding cut in π , since each block is an entire value interval. Thus σ is indecomposable. It cannot have size one: a one-block original permutation would be increasing, hence decomposable if its size exceeded one. Therefore $\sigma \in \mathcal{R}$.

Conversely, take $\sigma \in \mathcal{R}$ and replace each record by an arbitrary nonempty increasing interval block, leaving nonrecords as singleton blocks. The relative order of all blocks is prescribed by σ . A decreasing triple cannot use two points of an increasing block; if it uses three blocks it projects to a 321 in σ . Hence avoidance is preserved. The same block-order observation shows that every inflated record consists entirely of records, while an inflated nonrecord remains a nonrecord.

A hypothetical sum cut in the inflated permutation either lies between blocks or inside a block. In the former case it descends to a sum cut of σ . In the latter case, all earlier blocks must have smaller values than that block and all later blocks must have larger values. The underlying entry would then be a singleton direct-sum component of σ , also impossible in a nontrivial indecomposable permutation. Thus inflation preserves indecomposability.

Every bond internal to an inflated record block is an upper bond. No upper bond crosses two blocks: otherwise their underlying entries would already be adjacent consecutive records in σ . Consequently the inflated blocks are exactly the maximal record runs, together with the unchanged nonrecords. Contracting recovers σ and all chosen lengths. This proves existence, uniqueness, and the inverse property. $\square$

Example 5.2. The indecomposable permutation 235614 contracts to 2413. The record blocks are 23 and 56, while 1 and 4 are unchanged nonrecords. The core has block lengths (2, 2, 1, 1); inflating its two records by those lengths reconstructs 235614. In contrast, the consecutive lower run 123 in 4123 is not contracted.

A record of weight xy becomes a nonempty record run of weight

$$xy + (xy)^2 + (xy)^3 + \cdots = \frac{xy}{1 - xy}.$$

Theorem 5.1 therefore translates to

$$I(x, y) = R\left(x, \frac{y}{1 - xy}\right). \quad (17)$$

The inverse formal substitution is $y \mapsto y/(1 + xy)$, giving

$$\boxed{R(x, y) = I\left(x, \frac{y}{1 + xy}\right)}. \quad (18)$$

These substitutions are well-defined coefficientwise in x : all coefficients are polynomials in y , and the substituted series have invertible denominators with constant term 1. The alternating signs in the inverse substitution express an inversion of a proved bijection, not an unmotivated guess or an assumed inclusion–exclusion rule.

6 Derivation of the conjectured expression

Substitute (18) into (15) and clear the invertible denominator $1 + xy$. We obtain

$$(1 + xy)R(x, y)^2 + (x + x^2y - 1)R(x, y) + x^2y = 0. \quad (19)$$

Setting $y = 1$ gives

$$\boxed{(1 + x)R^2 + (x + x^2 - 1)R + x^2 = 0}. \quad (20)$$

Equivalently,

$$R = x^2 + (x + x^2)R + (1 + x)R^2. \quad (21)$$

This determines R uniquely from its zero constant term; coefficient recursion also gives $[x]R = 0$.

The discriminant of (20) is

$$(1 - x - x^2)^2 - 4x^2(1 + x) = D(x).$$

The correct root is therefore

$$R(x) = \frac{1 - x - x^2 - \sqrt{D(x)}}{2(1 + x)}. \quad (22)$$

The other root has constant term 1 and cannot count a class whose members have size at least two. The first core coefficients are

$$R = x^2 + x^3 + 3x^4 + 7x^5 + 19x^6 + 53x^7 + 153x^8 + 453x^9 + 1367x^{10} + 4191x^{11} + \cdots.$$

Substituting (22) into the combinatorial assembly formula (9) and simplifying gives (4). For clarity, the following polynomial certificates make this last algebraic step checkable without trusting symbolic simplification. Put

$$G(x, r) = (1 + x)r^2 + (x + x^2 - 1)r + x^2, \quad s = 1 - x - x^2 - 2(1 + x)r.$$

Then direct expansion gives

$$s^2 - D(x) = 4(1 + x)G(x, r), \quad (23)$$

$$(P - Qs)(1 - r)^2 - 8x(1 + x)^2(1 - r + x) = 2(Qr - 3x^2 - 7x - 3)G(x, r). \quad (24)$$

At $r = R(x)$, the right sides vanish, and s has constant term 1. Thus $s = \sqrt{D(x)}$ and (24) proves the desired identity with (9). This completes the proof of Theorem 1.1.

An additional compact algebraic certificate is

$$4x(1+x)^2A(x)^2 - P(x)A(x) + 2x^3 + 8x^2 + 9x + 3 = 0, \quad A(0) = 1. \quad (25)$$

The constant term condition and the nonzero coefficient $-P(0) = -3$ determine a unique formal solution. This identity follows either from (4) by squaring or from the two preceding polynomial certificates.

7 Refinements and explicit core coefficients

7.1 Records, nontrivial components, and the singleton

For $\pi \in \mathcal{A}$, let $m(\pi)$ count its nontrivial direct-sum components, and let $s(\pi) \in \{0, 1\}$ count its singleton components. The same unique decomposition used earlier gives the full refinement

$$\sum_{\pi \in \mathcal{A}} x^{|\pi|} y^{\text{rec}(\pi)} u^{m(\pi)} v^{s(\pi)} = \frac{1}{1 - uR(x, y)} + \frac{vxy}{(1 - uR(x, y))^2}. \quad (26)$$

Thus permutations with m nontrivial components and no singleton have generating function $R(x, y)^m$; those with one singleton have generating function $(m+1)xyR(x, y)^m$. No additional interaction term is needed, precisely because the boundary lemma has already handled possible bonds across components.

7.2 An alternating Narayana formula

Let $r_{n,k} = [x^n y^k]R(x, y)$. From (14), (13), and (18),

$$R(x, y) = \sum_{m \geq 2} \sum_{\ell \geq 1} N(m-1, \ell) x^m y^\ell (1+xy)^{-\ell}.$$

Using

$$(1+xy)^{-\ell} = \sum_{d \geq 0} (-1)^d \binom{\ell+d-1}{d} x^d y^d$$

and setting $m = n - d$, $\ell = k - d$ proves

$$r_{n,k} = \sum_{d=0}^{k-1} (-1)^d \binom{k-1}{d} N(n-d-1, k-d). \quad (27)$$

The convention after (13) handles all out-of-range indices. Although the sum alternates, its value is nonnegative because it counts the reduced cores just constructed. For example,

$$[x^4]R(x, y) = y + 2y^2, \quad [x^5]R(x, y) = y + 5y^2 + y^3.$$

7.3 A positive Catalan expansion

Equation (20) can instead be written

$$R = a + bR^2, \quad a = \frac{x^2}{1-x-x^2}, \quad b = \frac{1+x}{1-x-x^2}.$$

The solution with zero constant term is $a \sum_{j \geq 0} \text{Cat}_j(ab)^j$, where $\text{Cat}_j = \binom{2j}{j}/(j+1)$. Consequently

$$R(x) = \sum_{j \geq 0} \text{Cat}_j \frac{x^{2j+2}(1+x)^j}{(1-x-x^2)^{2j+1}}. \quad (28)$$

Every factor has nonnegative coefficients, and only finitely many summands contribute at any fixed degree. For a fully finite evaluation of the rational factors, for integers $h \geq 1$, $m \geq 0$ one may use

$$[x^m](1-x-x^2)^{-h} = \sum_{\ell=0}^{\lfloor m/2 \rfloor} \binom{h+m-\ell-1}{m-\ell} \binom{m-\ell}{\ell}. \quad (29)$$

Indeed, expand first in powers of $x+x^2$, then choose which of the $m-\ell$ factors contribute x^2 . Together with the finite expansion of $(1+x)^j$, this gives a nonalternating binomial formula for every r_n .

8 Exact coefficient algorithms and checks

8.1 Positive quadratic-arithmetic recurrences

Put $r_n = [x^n]R$ and $s_n = [x^n](1-R)^{-1}$. With negative-index coefficients zero, $r_0 = r_1 = 0$, and $s_0 = 1$, equation (21) gives

$$r_n = \mathbf{1}_{n=2} + r_{n-1} + r_{n-2} + \sum_{i+j=n} r_i r_j + \sum_{i+j=n-1} r_i r_j. \quad (30)$$

Every sum here uses nonnegative indices. Because $r_0 = r_1 = 0$, this recurrence is not circular. Next,

$$s_n = \sum_{k=2}^n r_k s_{n-k} \quad (n \geq 1), \quad (31)$$

$$a_0 = 1, \quad a_n = s_n + \sum_{j=0}^{n-1} s_j s_{n-1-j} \quad (n \geq 1). \quad (32)$$

These formulas calculate all terms through index N in $O(N^2)$ integer arithmetic operations. They use only addition and multiplication of nonnegative integers, making them useful as an independent check on a faster signed recurrence.

8.2 A linear-arithmetic coefficient algorithm

Let

$$T(x) = \sqrt{D(x)} = \sum_{n \geq 0} t_n x^n, \quad (d_0, d_1, d_2, d_3, d_4) = (1, -2, -5, -2, 1).$$

Differentiating $T^2 = D$ gives $2DT' = D'T$. Equating coefficients of x^{n-1} yields, for $n \geq 1$,

$$t_0 = 1, \quad 2n t_n = \sum_{k=1}^{\min(4,n)} (3k-2n) d_k t_{n-k}. \quad (33)$$

Now write $p_j = [x^j]P$, so $(p_0, \dots, p_4) = (3, 10, 11, 4, 1)$ and $p_j = 0$ for $j > 4$. Comparing coefficients of x^{n+1} in $8x(1+x)^2 A = P - QT$ gives

$$a_n = \frac{p_{n+1} - 3t_{n+1} - 5t_n - t_{n-1}}{8} - 2a_{n-1} - a_{n-2} \quad (n \geq 0), \quad (34)$$

where negative subscripts mean zero. Compute t_1 before using the formula at $n = 0$; it gives $t_1 = -1$ and $a_0 = 1$ as required.

All divisions are exact. For the t_n , this also follows directly from (22), which expresses T as the integer series $1 - x - x^2 - 2(1 + x)R$. For the division by 8, the proved identity for A and its combinatorial interpretation give integrality.

There are only four summands in (33) and a fixed number of operations in (34). Thus all terms through index N require $O(N)$ integer arithmetic operations. This is an arithmetic-operation bound, not a claim of $O(N)$ bit complexity. Section 9 shows that a_n has $\Theta(n)$ bits; the complete output has $\Theta(N^2)$ bits. A streaming implementation needs only a fixed number of recent large integers, hence $O(N)$ working bits apart from output. The supplied implementation retains arrays for convenient comparison.

8.3 Independent verification and reproducibility

The accompanying `verify.py` uses Python's standard library only. Its mesh predicate implements the rectangles directly and does not invoke Theorem 3.1. To enumerate $\text{Av}_n(321)$ efficiently, it inserts the new largest entry n into each smaller avoider exactly at positions whose suffix is increasing. This generator is correct because any new 321 must use n as its first entry; removing n gives the unique parent. As a separate check, the generated sets were compared with a filter of all $n!$ permutations through $n = 8$.

The completed verification run examined all 82,500 Catalan-class permutations of lengths 0 through 11. For every one, the direct mesh predicate agreed with the structural characterization; the two mesh counts agreed with a_n ; the record-to-Dyck bijection round-tripped; and the refined generating function (26) agreed with the joint counts. There were 23,713 contraction/inflation round trips on nontrivial indecomposables. A further 102 independently chosen record inflations of small cores also round-tripped. Pointwise transposition of mesh occurrences was checked through length 8.

n	$ \text{Av}_n(321) $	avoid (12, 174)	avoid (12, 234)	r_n
0	1	1	1	0
1	1	1	1	0
2	2	1	1	1
3	5	3	3	1
4	14	6	6	3
5	42	18	18	7
6	132	47	47	19
7	429	139	139	53
8	1430	405	405	153
9	4862	1225	1225	453
10	16796	3740	3740	1367
11	58786	11602	11602	4191

Table 1: Exhaustive geometric enumeration, not an expansion of the conjectured formula. The final column counts nontrivial reduced indecomposable cores.

The positive recurrences and the linear-arithmetic recurrence agree at all 1001 indices $0 \leq n \leq 1000$, including all 17 terms currently listed in the consulted OEIS entry. For orientation, the extension begins

n	a_n	n	a_n
17	12380078	22	4778632088
18	40423803	23	15884348886
19	132564051	24	52941973417
20	436417933	25	176889898401
21	1441809621	26	592374142673

These tests guard against errors in the shading convention, the bijections, the refinements, and the algebra. They are not the proof of an identity for all n ; the structural constructions and formal derivations above are.

9 Asymptotic growth

9.1 The dominant singularity

The discriminant is reciprocal. For $x \neq 0$,

$$x^{-2}D(x) = (x + x^{-1})^2 - 2(x + x^{-1}) - 7. \quad (35)$$

Hence its roots satisfy $x + x^{-1} = 1 \pm 2\sqrt{2}$. The plus sign yields a positive reciprocal pair. Define

$$\alpha = \frac{1 + 2\sqrt{2} + \sqrt{5 + 4\sqrt{2}}}{2} = 3.546455444684995244456761\dots, \quad (36)$$

$$\rho = \alpha^{-1} = 0.281971680061194853146616\dots \quad (37)$$

The minus sign gives two distinct roots on the unit circle, because $-2 < 1 - 2\sqrt{2} < 2$. Thus ρ is the unique smallest-modulus zero of D ; it is positive and simple. The denominator of (4) has no other singularity of modulus at most ρ , with the point at zero already known to be removable. Since $Q(\rho) > 0$, the square-root term does not cancel. Therefore the radius of convergence is exactly ρ .

Put

$$B(x) = \frac{Q(x)}{8x(1+x)^2}, \quad \kappa = \sqrt{-\rho D'(\rho)}. \quad (38)$$

In a slit neighborhood of ρ , compatible with the branch from the origin,

$$\sqrt{D(x)} = \kappa(1 - x/\rho)^{1/2} + O((1 - x/\rho)^{3/2}).$$

The leading nonanalytic term in A is therefore $-B(\rho)\kappa(1 - x/\rho)^{1/2}$. The coefficient estimate

$$[x^n](1 - x/\rho)^{1/2} \sim -\frac{\rho^{-n}}{2\sqrt{\pi}n^{3/2}}$$

follows from the binomial coefficients and Stirling's formula. Standard singularity transfer applies here because this algebraic function has a unique dominant branch point and admits analytic continuation to a dented disk beyond ρ [4].

Theorem 9.1 (Asymptotic enumeration). *With α and ρ given by (36)–(37),*

$$a_n \sim K\alpha^n n^{-3/2}, \quad K = \frac{Q(\rho)\sqrt{-\rho D'(\rho)}}{16\sqrt{\pi}\rho(1+\rho)^2} = 0.413931068036970644808821\dots \quad (39)$$

More precisely,

$$a_n = K\alpha^n n^{-3/2} \left(1 + \frac{c_1}{n} + O(n^{-2}) \right), \quad (40)$$

where

$$c_1 = \frac{3}{8} + \frac{3\rho B'(\rho)}{2B(\rho)} + \frac{3\rho D''(\rho)}{8D'(\rho)} = -1.008253037811239547502410\dots \quad (41)$$

Proof. Only the correction term remains to be derived. Set $z = 1 - x/\rho$. Taylor expansion gives

$$D(x) = -\rho D'(\rho)z + \frac{\rho^2 D''(\rho)}{2}z^2 + O(z^3),$$

and therefore

$$\sqrt{D(x)} = \kappa z^{1/2} \left(1 - \frac{\rho D''(\rho)}{4D'(\rho)}z + O(z^2) \right).$$

Also $B(x) = B(\rho) - \rho B'(\rho)z + O(z^2)$. The ratio of the $z^{3/2}$ singular coefficient of A to its $z^{1/2}$ coefficient is

$$-\frac{\rho B'(\rho)}{B(\rho)} - \frac{\rho D''(\rho)}{4D'(\rho)}.$$

The $z^{1/2}$ transfer has its own relative correction $3/(8n)$. The ratio of the leading $z^{3/2}$ transfer constant to the $z^{1/2}$ constant is $\Gamma(-1/2)/\Gamma(-3/2) = -3/2$. Combining the two contributions yields (41). Higher local terms give a relative $O(n^{-2})$ error; the other singularities have larger modulus. $\square$

For a numerical check, the table shows the exact coefficient divided by its leading asymptotic estimate. The last column should converge to c_1 .

n	$a_n/(K\alpha^n n^{-3/2})$	$n(a_n/(K\alpha^n n^{-3/2}) - 1)$
25	0.961218942324	-0.969526441895
50	0.980230192460	-0.988490377020
100	0.990017308751	-0.998269124946
250	0.995983062048	-1.004234488091
500	0.997987520864	-1.006239567862
1000	0.998992754749	-1.007245251130

At $n = 1000$, the exact coefficient divided by the two-term estimate is $1.000001008804\dots$, consistent with the proved relative $O(n^{-2})$ remainder.

Since $|\text{Av}_n(321)| = \text{Cat}_n \sim 4^n/(\sqrt{\pi}n^{3/2})$, the proportion of 321-avoiders satisfying the mesh restriction is

$$\frac{a_n}{\text{Cat}_n} \sim K\sqrt{\pi} \left(\frac{\alpha}{4} \right)^n. \quad (42)$$

Thus the additional geometric restriction is exponentially selective, while retaining the characteristic square-root exponent $n^{-3/2}$.

9.2 A limiting probability for the singleton component

The decomposition also gives a statistic unavailable from an unstructured fit to initial coefficients. Let π be uniformly distributed among the length- n members of $\mathcal{A}$. Define

$$R_0 = R(\rho) = \frac{1 - \rho - \rho^2}{2(1 + \rho)} = 0.249038376398374331486125\dots$$

The two summands of (9) count zero and one singleton, respectively. Near the dominant singularity,

$$R(x) = R_0 - \frac{\kappa}{2(1+\rho)}(1-x/\rho)^{1/2} + O(1-x/\rho).$$

The leading singular amplitudes of the two summands are proportional to $(1-R_0)^{-2}$ and $2\rho(1-R_0)^{-3}$. Taking their ratio proves

$$\lim_{n \rightarrow \infty} \Pr(s(\pi) = 1) = \frac{2\rho}{1-R_0+2\rho} = 0.428885255667078255465240\dots \quad (43)$$

The denominators are nonzero because $R_0 < 1$; the same singularity-transfer argument applies to both summands.

10 Conclusion

The conjecture follows from a sequence of exact, reversible constructions. The forbidden mesh first separates into a record-bond obstruction and a pair-of-singletons obstruction. Direct-sum decomposition removes the latter; canonical record-run contraction handles the former. The remaining record-refined enumeration is Narayana, whose equation is obtained from an explicit Dyck-path bijection. These steps explain both the discriminant $D(x)$ and the rational prefactor in the conjectured expression.

The result is not only an algebraic generating function. The same proof provides a joint statistic refinement, explicit core coefficients, two exact coefficient algorithms, and asymptotic constants. The supplied computations are independent checks of the delicate constructions, while the proof of all coefficients is entirely in the preceding combinatorial arguments.

A Contents and reproduction of the archive

The archive contains the source `article.tex`, the compiled `article.pdf`, `verify.py`, and the optional `symbolic_check.py`. The `data/` directory contains coefficients through index 1000, exhaustive count tables, exact algebraic certificates, asymptotic constants and ratio checks, and a machine-readable verification report. Logs record the completed runs. The extended b-file is generated here and is not represented as an official OEIS data file.

From the extracted directory, run

```
python verify.py --max-n 1000 --exhaustive 11
python symbolic_check.py
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The second command requires SymPy and mpmath; it is optional because the polynomial certificates are displayed explicitly in the article. The main verification uses no third-party package. Python 3.9 or later is sufficient; the recorded run used Python 3.13.5, SymPy 1.14.0 and mpmath 1.3.0. Do not invoke the checker with Python's `-O` flag, which disables its assertions. Increasing the exhaustive limit entails Catalan growth and is unrelated to the linear-arithmetic cost of generating coefficients.

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